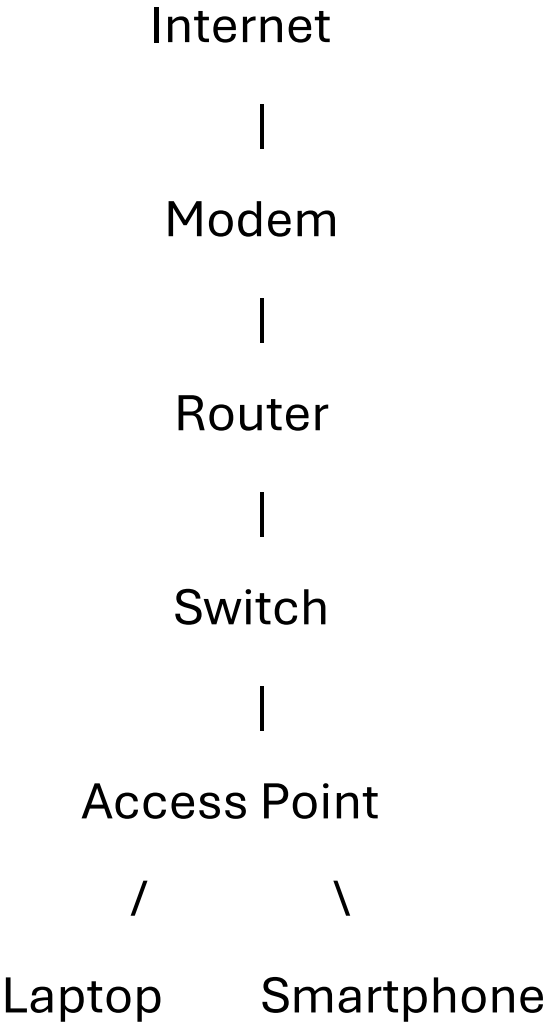


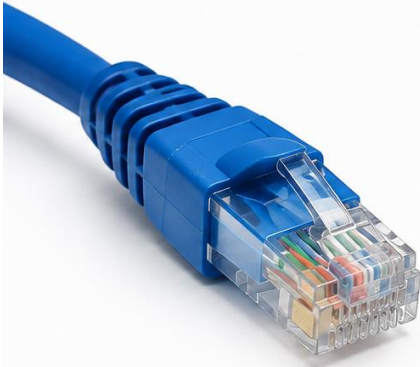
Task 1: Typical Home Wi-Fi Network Diagram



Task 2: Functions and Real-Life Examples

Device	Main Function	Real-Life Example
Modem	Connects the local network to the Internet Service Provider (ISP).	Used at home to receive Internet service from the ISP.
Router	Routes data between networks and shares the Internet connection with multiple devices.	Used at home to connect phones, laptops and other devices to the Internet.
Switch	Connects multiple wired devices on the same local network.	Used in an office to connect computers and printers using Ethernet cables.
Access Point	Provides wireless network access and extends Wi-Fi coverage.	Used in a large office or home to provide Wi-Fi in areas with weak signal.
Network Cable	Carries network data between wired networking devices.	Ethernet cable used to connect a PC to a router or switch.

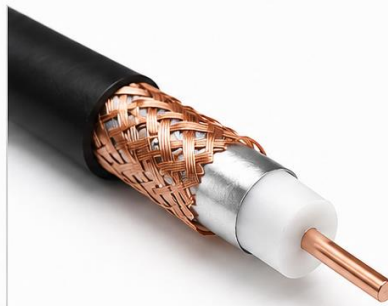
Types of Network Cables



1 Ethernet Cable

Common Use:

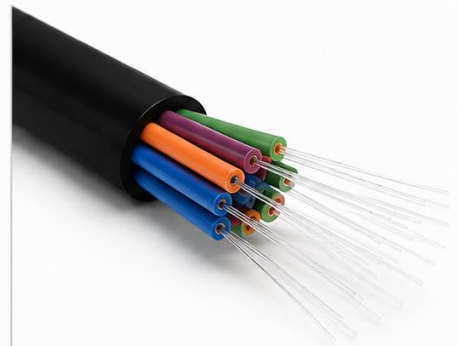
Used to connect computers and network devices in a wired LAN.



2 Coaxial Cable

Common Use:

Commonly used for cable TV and broadband Internet.



3 Fiber Optic Cable

Common Use:

Commonly used for high-speed and long-distance network connections.

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Ethernet Cable —

https://en.wikipedia.org/wiki/Ethernet_physical_layer

Coaxial Cable —

https://en.wikipedia.org/wiki/Coaxial_cable

Fiber Optic Cable —

https://en.wikipedia.org/wiki/Optical_fiber

Task 4: Router vs Switch Comparison

Difference	Router	Switch
Main Use	Connects different networks and shares Internet.	Connects devices within the same local network (LAN).
Typical Location	Commonly used at home, offices, and network gateways.	Commonly used inside offices, schools, and data centers.
Data Handling	Forwards data between different networks using IP addresses.	Forwards data between devices using MAC addresses.
Internet Connection	Usually connects the local network to the Internet.	Does not normally connect directly to the Internet.
Ports	Usually has fewer ports for connecting network devices.	Usually has many Ethernet ports for connecting multiple wired devices.

